THE TWO SPECTRAL SEQUENCES OF A CARTAN-EILENBERG RESOLUTION COMPUTE R^QF(K^P) AND R^PF(H^Q(K))

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Editorial note. Counted span is the author's own proof environment, source lines 7415-7468, following the statement at 7393-7413.

Editorial note. The three opening remarks are stated as 'used without further mention' but are explicitly listed with their inputs (7416-7430), recorded.

Allowed standard prerequisites (editorial).

- Total complex of a Cartan-Eilenberg resolution. homology-lemma-double-complex-gives-resolution, cross-chapter citation at proof line 7425 (tags mechanism). Supplies that Tot(I) is an injective resolution of K, so RF is defined at K with value F(Tot(I)).
- Spectral sequences of a double complex. homology-lemma-ss-double-complex and homology-lemma-first-quadrant-ss, cross-chapter citations at proof lines 7433 and 7436 (tags mechanism). Supplies the two spectral sequences, their boundedness, convergence and finite filtrations.

Source reference. Original reference 0133: focused-seeds/geometry/S9-der-two-ss-complex-functor/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0132: focused-seeds/geometry/S9-der-two-ss-complex-functor/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0130: focused-seeds/geometry/S9-der-two-ss-complex-functor/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor of abelian categories. Let $K^\bullet$ be a bounded below complex of $\mathcal{A}$. Let $I^{\bullet,\bullet}$ be a Cartan-Eilenberg resolution for $K^\bullet$. The spectral sequences $({}'E_r, {}'d_r)_{r\geq 0}$ and $({}''E_r, {}''d_r)_{r\geq 0}$ associated to the double complex $F(I^{\bullet,\bullet})$ satisfy the relations*

$$ {}'E_1^{p,q} = R^q F(K^p) \quad \text{and} \quad {}''E_2^{p,q} = R^p F(H^q(K^\bullet)) $$

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Moreover, these spectral sequences are bounded, converge to $H^*(RF(K^\bullet))$, and the associated induced filtrations on $H^n(RF(K^\bullet))$ are finite.

Proof. We will use the following remarks without further mention:

(1) As $I^{p,\bullet}$ is an injective resolution of K^p we see that RF is defined at $K^p[0]$ with value $F(I^{p,\bullet})$.
(2) As $H^p_I(I^{\bullet,\bullet})$ is an injective resolution of $H^p(K^\bullet)$ the derived functor RF is defined at $H^p(K^\bullet)[0]$ with value $F(H^p_I(I^{\bullet,\bullet}))$.
(3) By Homology, Lemma 0133 the total complex $\mathrm{Tot}(I^{\bullet,\bullet})$ is an injective resolution of $K^\bullet$. Hence RF is defined at $K^\bullet$ with value $F(\mathrm{Tot}(I^{\bullet,\bullet}))$.

Consider the two spectral sequences associated to the double complex $L^{\bullet,\bullet} = F(I^{\bullet,\bullet})$, see Homology, Lemma 0130. These are both bounded, converge to $H^*(\mathrm{Tot}(L^{\bullet,\bullet}))$, and induce finite filtrations on $H^n(\mathrm{Tot}(L^{\bullet,\bullet}))$, see Homology, Lemma 0132. Since $\mathrm{Tot}(L^{\bullet,\bullet}) = \mathrm{Tot}(F(I^{\bullet,\bullet})) = F(\mathrm{Tot}(I^{\bullet,\bullet}))$ computes $RF(K^\bullet)$ we find the final assertion of the lemma holds true.

Computation of the first spectral sequence. We have ${}'E_1^{p,q} = H^q(L^{p,\bullet})$ in other words

$$ {}'E_1^{p,q} = H^q(F(I^{p,\bullet})) = R^qF(K^p) $$

as desired. Observe for later use that the maps ${}'d_1^{p,q} : {}'E_1^{p,q} \to {}'E_1^{p+1,q}$ are the maps $R^qF(K^p) \to R^qF(K^{p+1})$ induced by $K^p \to K^{p+1}$ and the fact that R^qF is a functor.

Computation of the second spectral sequence. We have ${}''E_1^{p,q} = H^q(L^{\bullet,p}) = H^q(F(I^{\bullet,p}))$. Note that the complex $I^{\bullet,p}$ is bounded below, consists of injectives, and moreover each kernel, image, and cohomology group of the differentials is an injective object of $\mathcal{A}$. Hence we can split the differentials, i.e., each differential is a split surjection onto a direct summand. It follows that the same is true after applying F. Hence ${}''E_1^{p,q} = F(H^q(I^{\bullet,p})) = F(H^q_I(I^{\bullet,p}))$. The differentials on this are $(-1)^q$ times F applied to the differential of the complex $H^p_I(I^{\bullet,\bullet})$ which is an injective resolution of $H^p(K^\bullet)$. Hence the description of the E_2 terms. $\square$